Symbiosis Institute of Technology , Nagpur

AI-Powered Insurance Claim Automation System

From Manual to Magical: AI-Powered Claim Transformation

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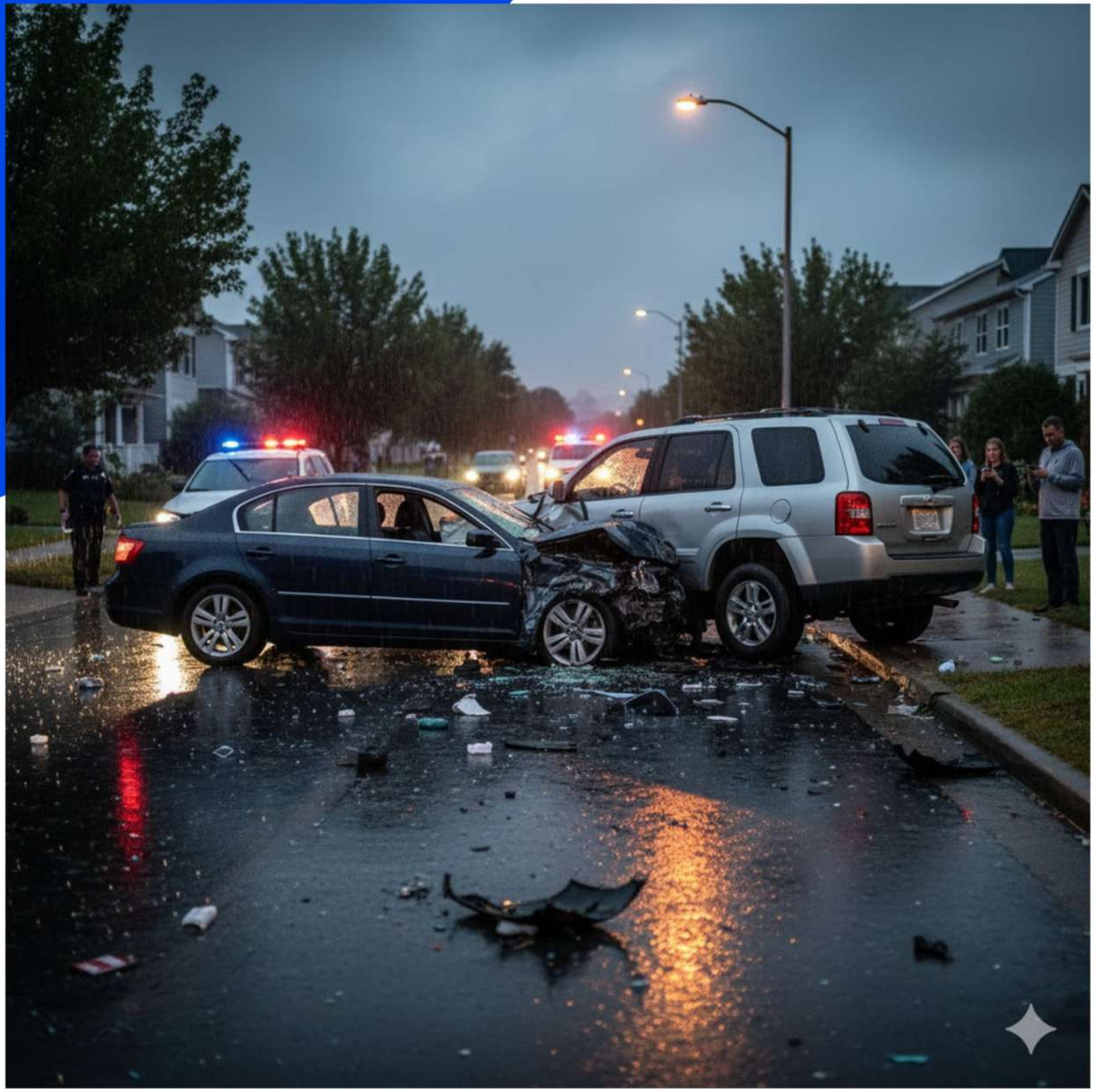
Introduction



The insurance sector handles thousands of claim requests daily.

Manually checking and approving claims is time-consuming and error-prone.

With Artificial Intelligence (AI), this process can be made faster, smarter, and more accurate.

The proposed system uses AI and Machine Learning to:

- Read and classify claim documents automatically.
- Predict whether a claim should be approved or denied.
- Provide clear, human-understandable explanations for every decision.
- This improves efficiency, transparency, and customer trust in the insurance process.

Problem statement

In the insurance industry, large volumes of claim-related documents such as claim forms, invoices, and reports are processed manually.

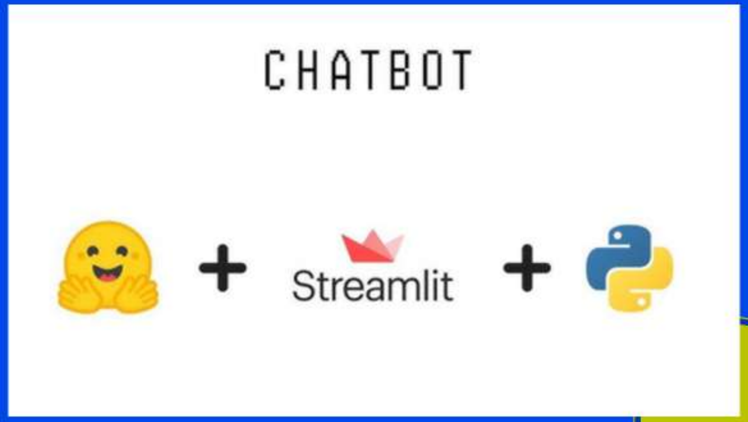
This manual process is slow, error-prone, and lacks transparency.

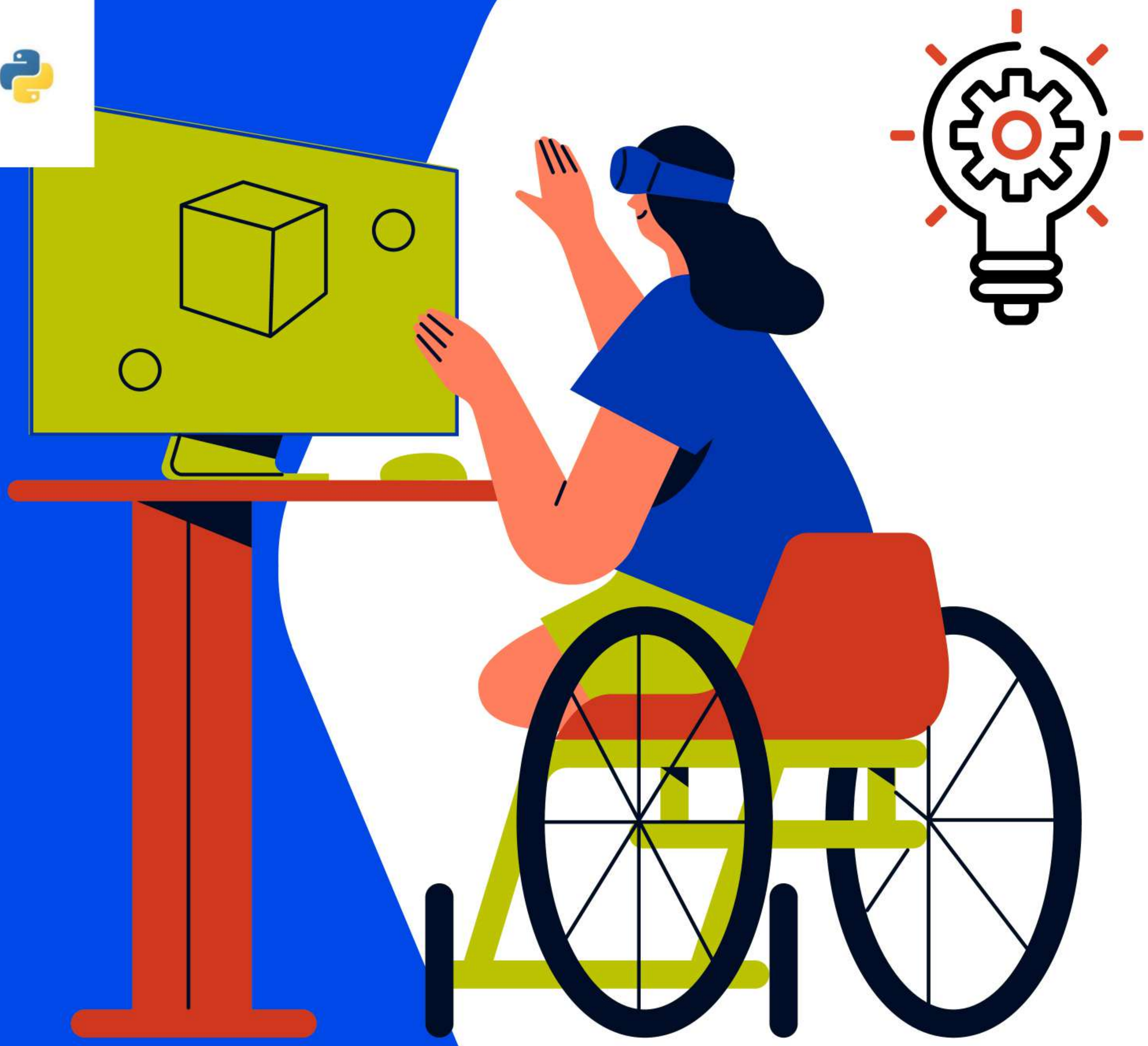
Customers often don’t understand why their claims were approved, reduced, or denied.

Key Issues:

⏳ Time-consuming claim assessment
⚙️ Manual document classification
💸 High operational costs
❓ Lack of clarity in decision-making

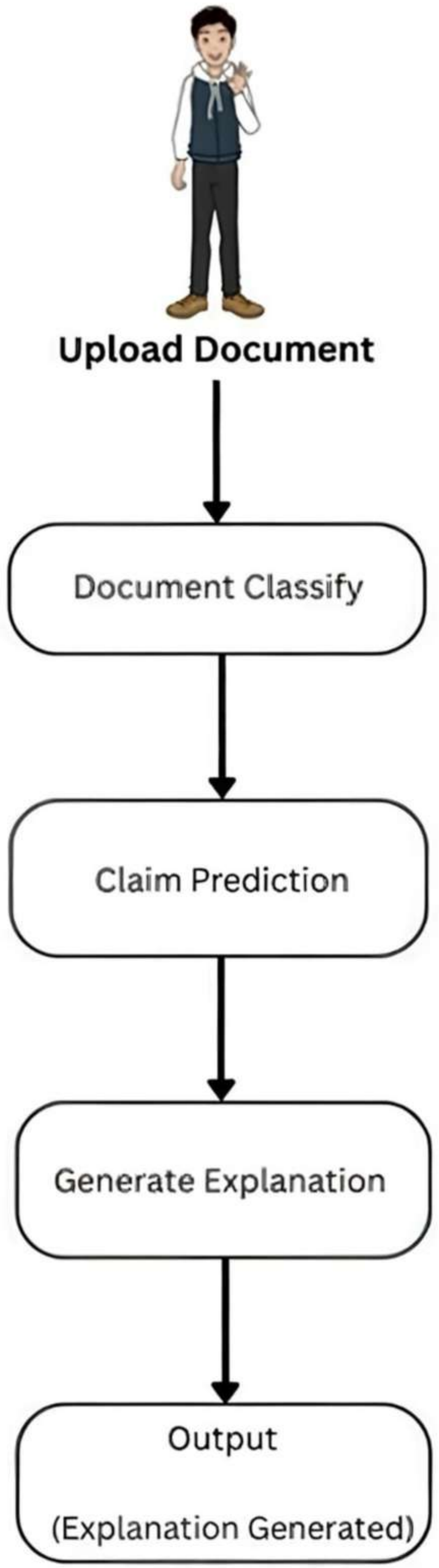
Objective





- Automate classification of insurance-related documents using AI.
- Predict claim outcomes (Approved / Denied) based on policy and vehicle data
- Use Generative AI to explain claim decisions in clear, human-friendly language.
- Build an interactive dashboard to integrate all functionalities.

🧭 Goal: To reduce manual intervention, improve accuracy, and enhance transparency in the insurance claim process.



How it works?

Stage	Description	Technology
Document Classification	Reads uploaded PDF, classifies as Claim Form / Invoice / Report	BART / Zero-Shot Transformer
Claim Prediction	Predicts claim approval or denial using policy data	Random Forest / XGBoost
Explanation Generation	Explains reasoning in plain English	FLAN-T5 / Gemini (Generative AI)
Dashboard	Combines all modules in one UI	Streamlit + Python

Tools and Technologies Used

Category	Tools / Libraries
Programming Language	Python
AI / ML Libraries	scikit-learn, transformers, pandas
Generative AI	FLAN-T5, Gemini API
NLP	pdfplumber, Zero-shot BART
Deployment	Streamlit, Ngrok
Environment	Google Colab / VS Code

Skills Utilized

🤖 Machine Learning (Random Forest, Model Training)

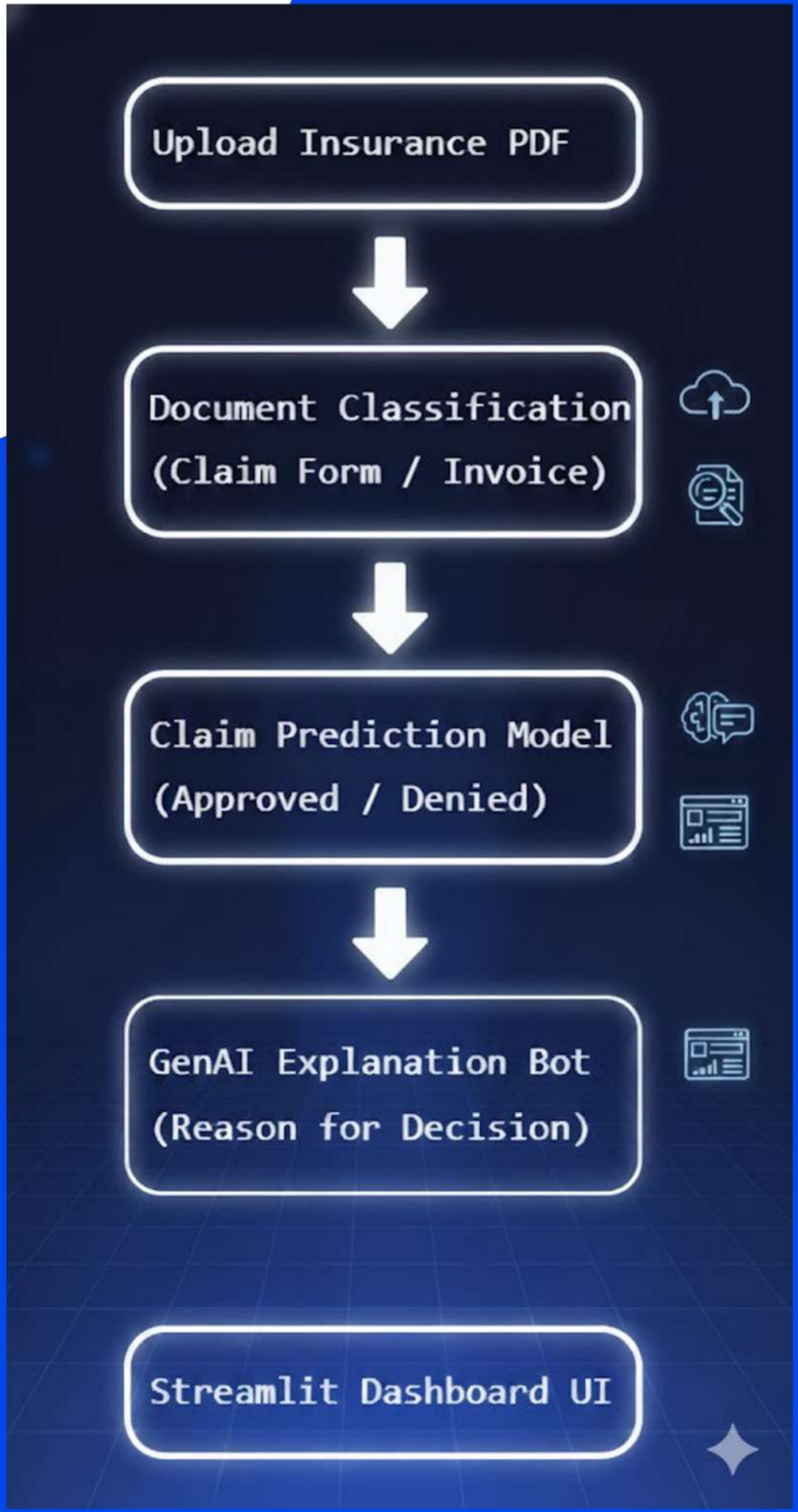
🧠 NLP (Transformer-based text understanding)

💬 Generative AI (Text generation and explanation)

💻 Python Development

📊 Data Analysis and Visualization

🌐 Streamlit Web App Deployment



System Flow Diagram

Step-by-step approach:

🧾 Upload PDF document
→ Extract text using pdfplumber or OCR

🤖 Document Classification
→ Identify type (Invoice / Claim Form / Report)

🧠 Claim Decision Prediction
→ Analyze features (Vehicle Age, Safety Rating, etc.)

→ Predict Approved or Denied
💬 Generative AI Explanation
→ Generate clear natural language explanation

📊 Streamlit Dashboard
→ Display results interactively with confidence score

Stage 1 – Document Classification Agent

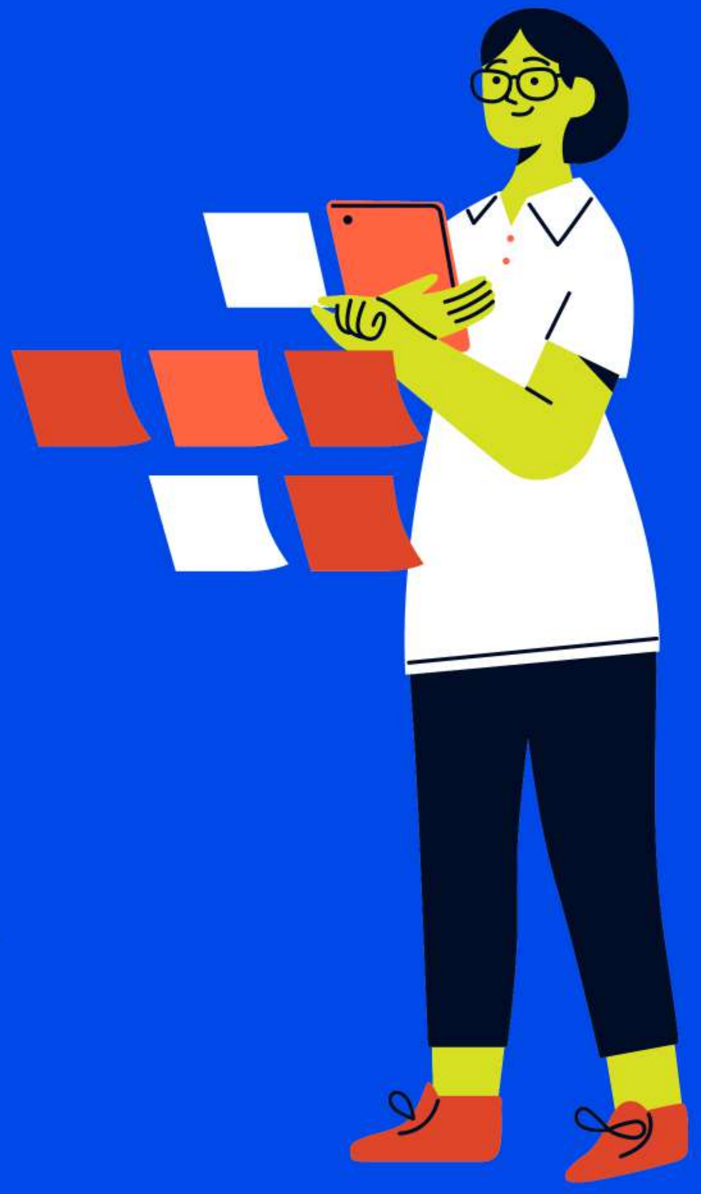
Purpose: Identify the type of uploaded document.

How it works:

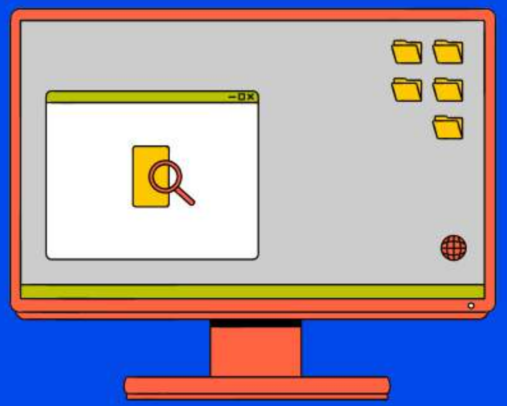
- Reads the text from PDFs (using OCR / pdfplumber).
- Uses an AI model (like Zero-shot BERT) to understand meaning.
- Classifies document as Invoice, Claim Form, Inspection Report, etc.
- Example:

"AUTO SERVICE INVOICE" → Invoice (96% confidence)

- Technologies Used:
- pdfplumber / PyMuPDF → Extract text from PDF files.
- Tesseract OCR → Read scanned or image-based documents.
- Transformers Library (Hugging Face) → For AI models like Zero-Shot BERT.
- Zero-Shot BERT Model → Classifies documents without needing labeled training data.
- Pandas / NumPy → For data handling and preprocessing.



Stage 2 – Claim Decision Predictor



Purpose: Predict whether the claim should be Approved or Denied.

Input Data:

Vehicle age

Safety rating (NCAP)

Customer age

Safety features (Airbags, Brake Assist, etc.)

Model Used: Random Forest Classifier

Example:

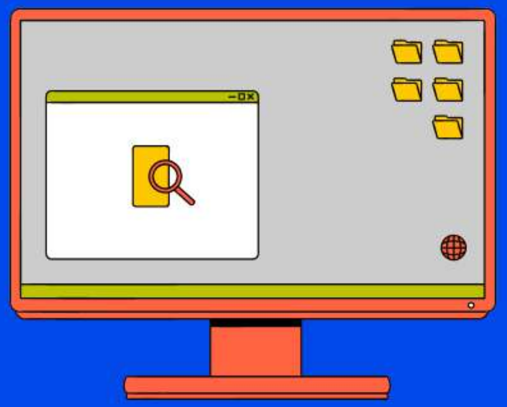
Vehicle age = 2 years, NCAP = 5 → ✅
Claim Approved (91%)

Vehicle age = 8 years, NCAP = 2 → ❌
Claim Denied (87%)

- Technologies Used:

- Scikit-learn (sklearn) → For building and training ML models.
- Random Forest Classifier → Predicts claim status based on input features.
- Matplotlib / Seaborn → For data visualization and analysis.
- CSV / Excel Dataset → Historical insurance data used for model training.

Stage 3 – AI Explanation Bot



Purpose: Explain the decision in human-understandable language.

Examples:

✅ “Your claim was approved because your vehicle is new and has high safety ratings.”

❌ “Your claim was denied because the vehicle is old and lacks important safety systems.”

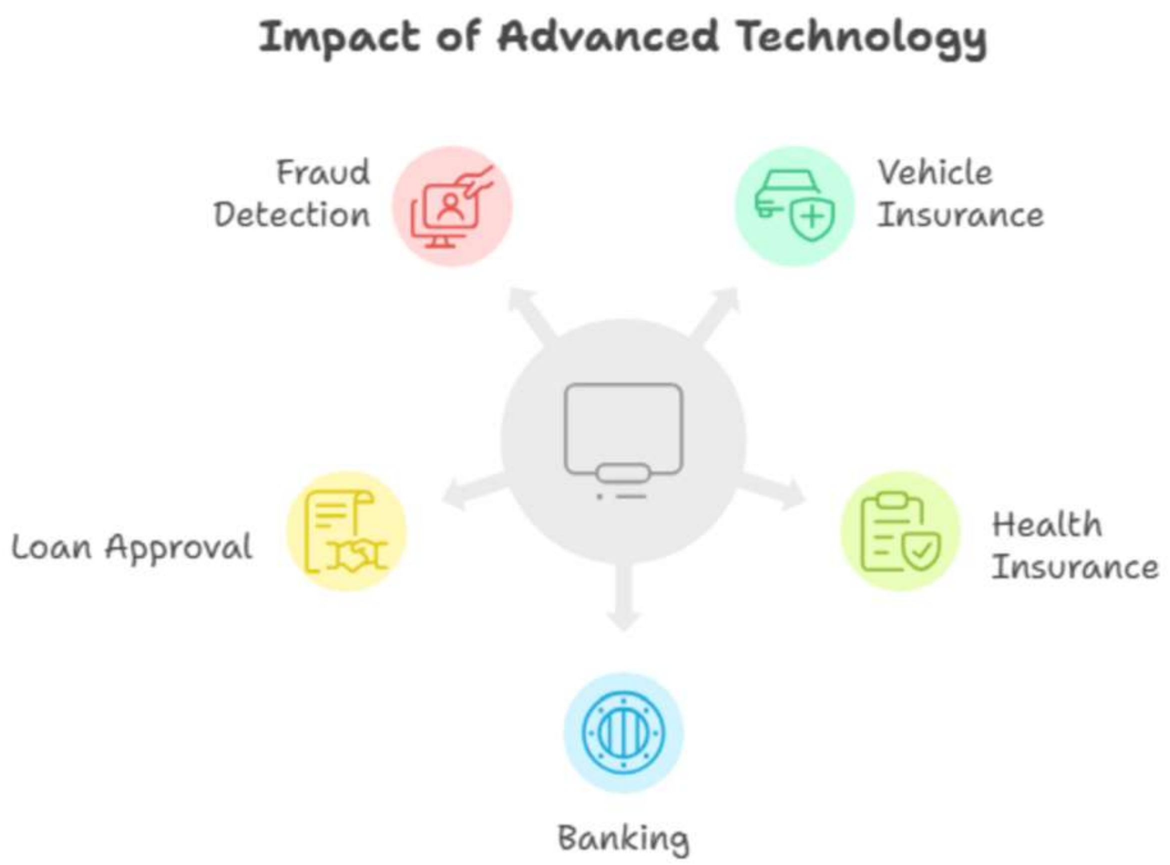
Benefit: Transparency and trust between customer and company.

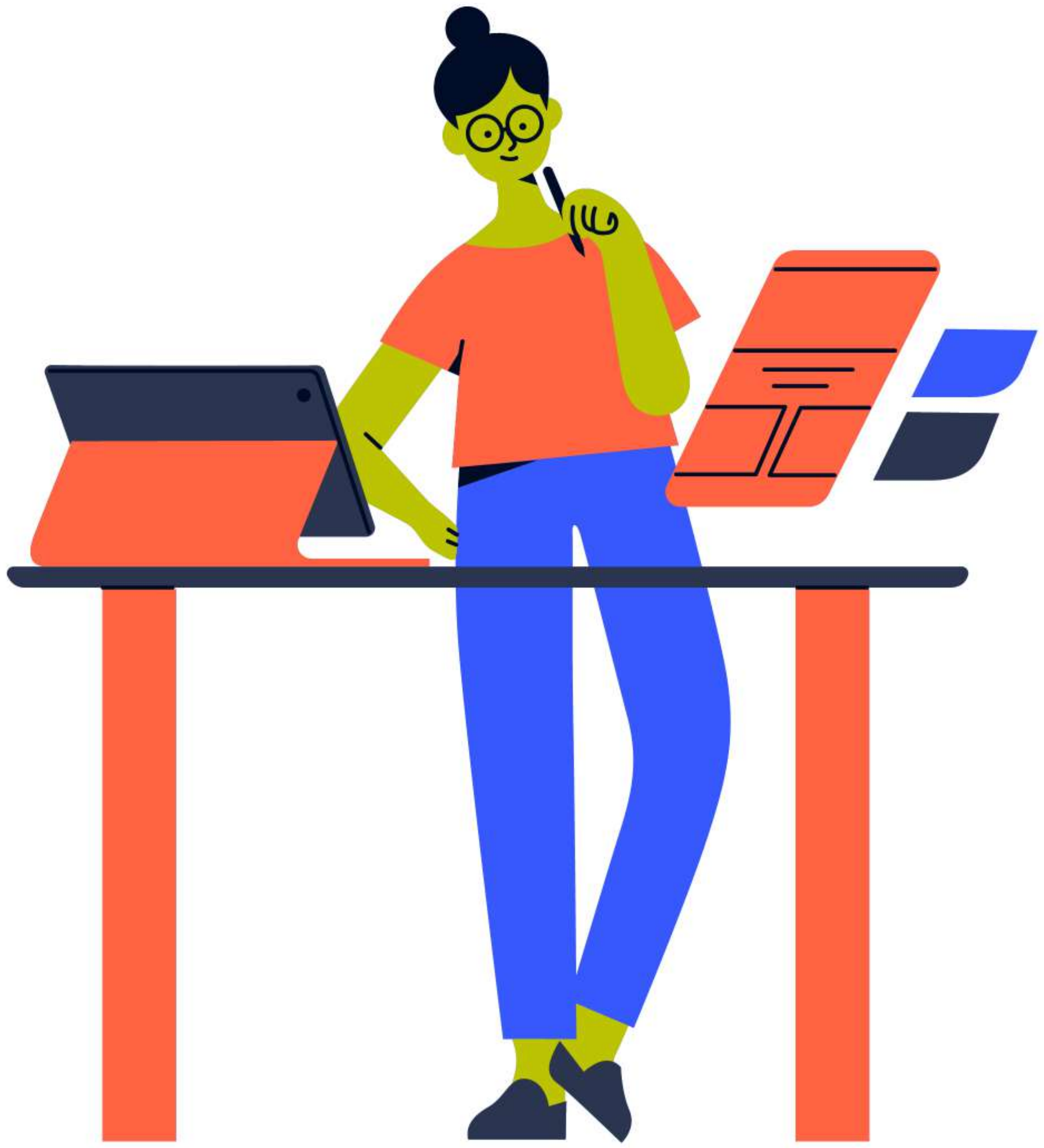
Technologies Used:

Natural Language Processing (NLP) → To generate easy-to-understand text.

GPT-based Language Model / LLM → Writes human-like explanations.

Python Functions & Conditional Logic → Combine prediction results with explanation rules.





Applications

- Vehicle and health insurance claim processing
- Banking document verification
- Automated loan approval systems
- Fraud detection in financial services

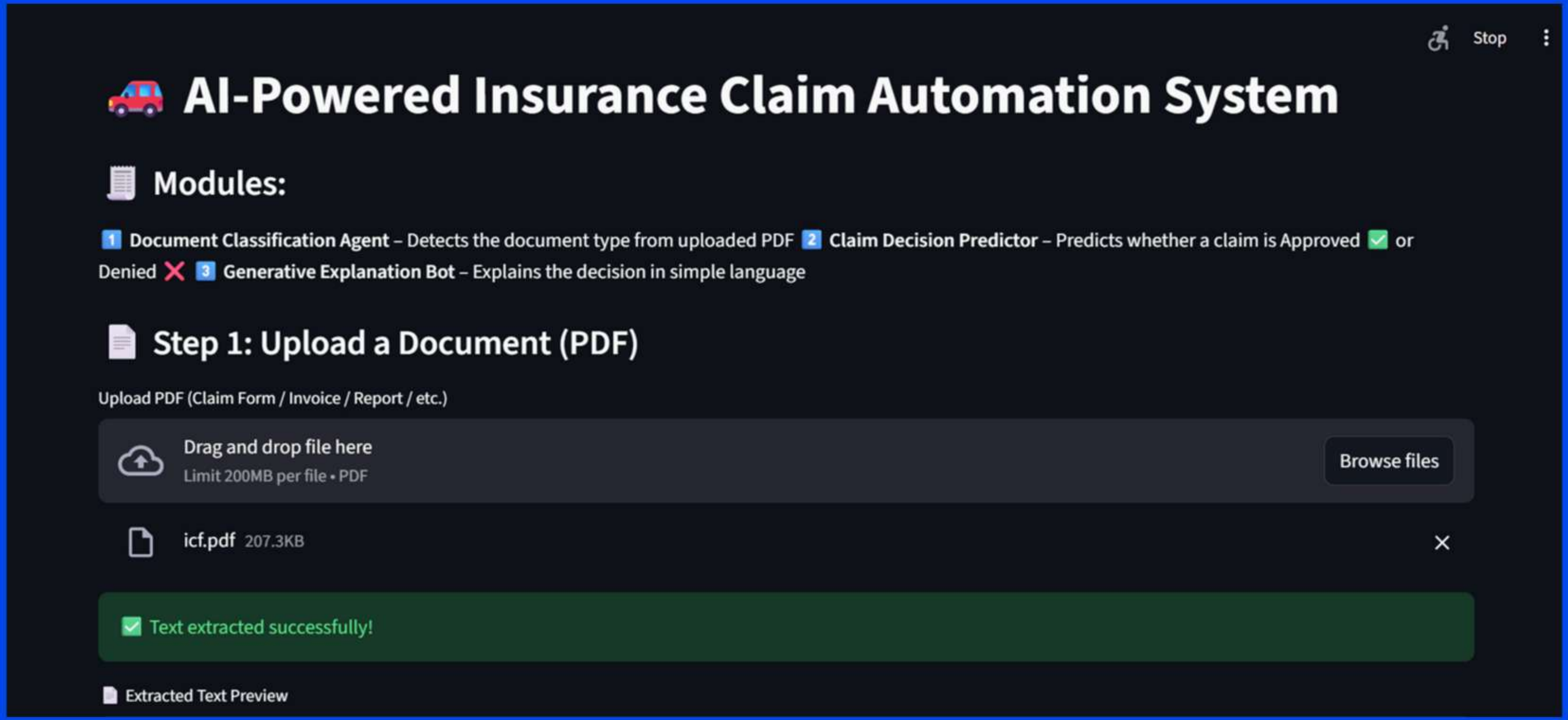
Future Scope

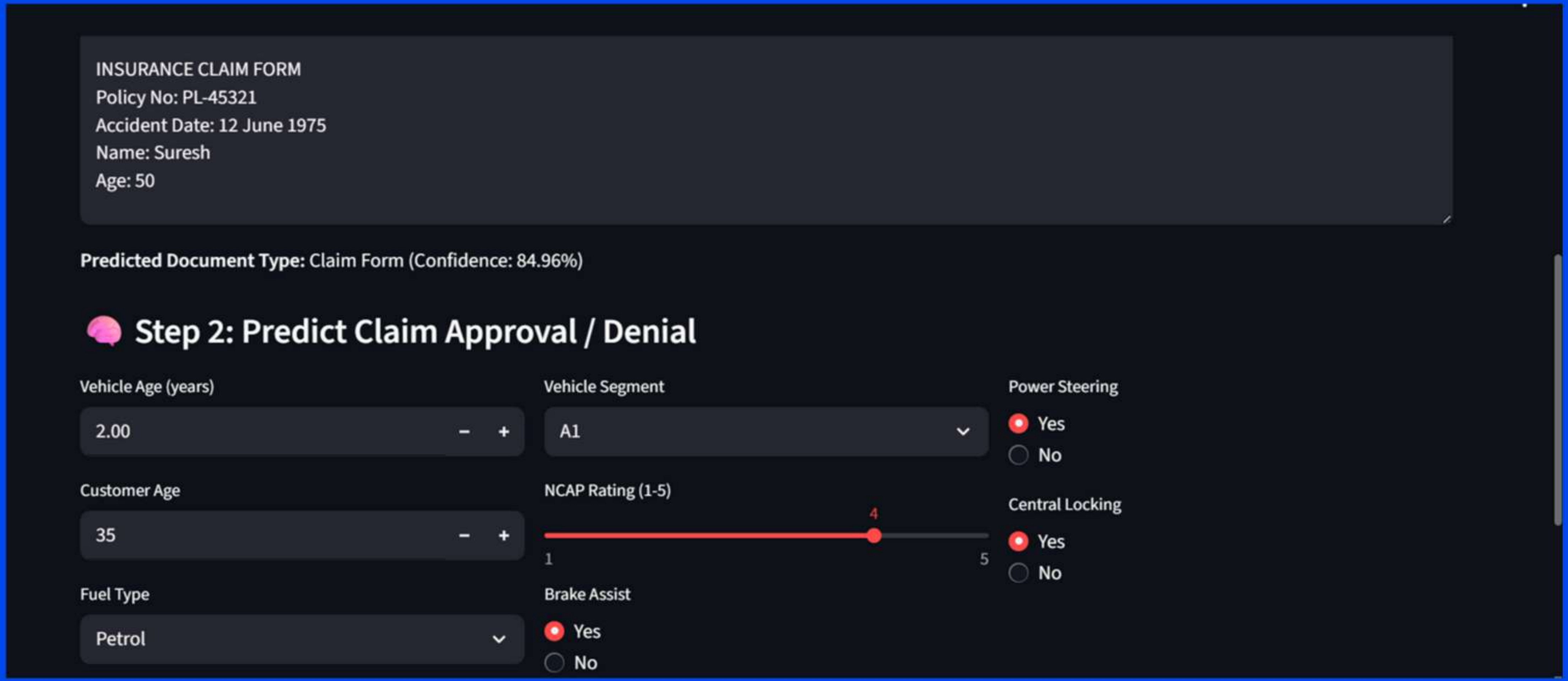
- Integrate Chatbot for real-time customer query handling
- Add Image-based damage detection using CNN models
- Use Deep Learning for even higher accuracy
- Connect with Blockchain for secure record management

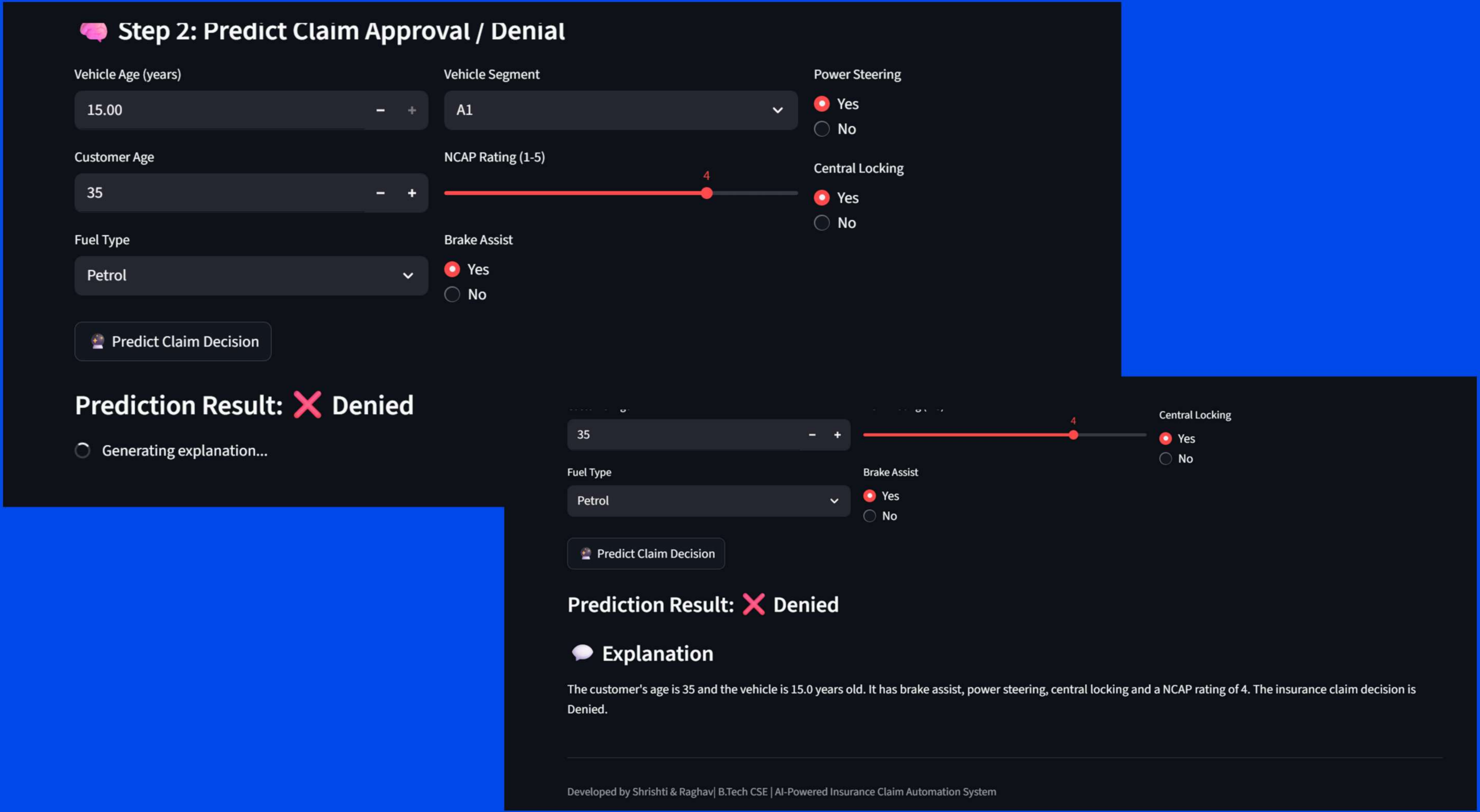
Conclusion

- The project shows how AI can revolutionize insurance workflows.
- It reduces manual effort and improves decision accuracy.
- The combination of document AI + prediction model + explanation bot makes the system intelligent and user-friendly.
- Truly, this system is "Shaping the Future of Smart and Transparent Insurance."

Dashboard


Stop
🚗 AI-Powered Insurance Claim Automation System
🧾 Modules:
1️⃣ Document Classification Agent – Detects the document type from uploaded PDF 2️⃣ Claim Decision Predictor – Predicts whether a claim is Approved ✅ or Denied ❌ 3️⃣ Generative Explanation Bot – Explains the decision in simple language
📄 Step 1: Upload a Document (PDF)
Upload PDF (Claim Form / Invoice / Report / etc.)
Drag and drop file here
Limit 200MB per file • PDF
Browse files
icf.pdf 207.3KB
✅ Text extracted successfully!
📄 Extracted Text Preview


INSURANCE CLAIM FORM
Policy No: PL-45321
Accident Date: 12 June 1975
Name: Suresh
Age: 50
Predicted Document Type: Claim Form (Confidence: 84.96%)
🧠 Step 2: Predict Claim Approval / Denial
Vehicle Age (years)
2.00
Vehicle Segment
A1
Power Steering
Yes
No
Customer Age
35
NCAP Rating (1-5)
4
1
5
Central Locking
Yes
No
Fuel Type
Petrol
Brake Assist
Yes
No



Step 2: Predict Claim Approval / Denial
Vehicle Age (years)
15.00
Vehicle Segment
A1
Power Steering
Yes
No
Customer Age
35
NCAP Rating (1-5)
4
Central Locking
Yes
No
Fuel Type
Petrol
Brake Assist
Yes
No
Predict Claim Decision
Prediction Result: ❌ Denied
Generating explanation...
35
4
Central Locking
Yes
No
Fuel Type
Petrol
Brake Assist
Yes
No
Predict Claim Decision
Prediction Result: ❌ Denied
Explanation
The customer's age is 35 and the vehicle is 15.0 years old. It has brake assist, power steering, central locking and a NCAP rating of 4. The insurance claim decision is Denied.
Developed by Shrishti & Raghav| B.Tech CSE | AI-Powered Insurance Claim Automation System